Lecture 12

Last time

- Reaching a critical point
- PL is back
- Variance Reduction

Today

- Analyzing variance reduction.

Where did we leave off?

We were considering a finite sum

$$f(w) = \frac{1}{n} \sum_{i=1}^{n} f_i(w).$$

Given a snapshot $\tilde{w}$, we introduced

$$g(w, i; \tilde{w}) = \nabla f(\tilde{w}) + \nabla f_i(w) - \nabla f_i(\tilde{w}).$$

Last time, we saw this is an unbiased estimator and we propose the outer iteration:

(SVRG)

Set $\tilde{w}_0 = w_k$

Compute $\nabla f(\tilde{w}_0)$

For $t = 0, \ldots, T-1$

Draw a fresh $j \sim \text{Unif}[n]$

$$\tilde{w}_{t+1} \leftarrow \tilde{w}_t - \eta\, g(\tilde{w}_t, i_t; \tilde{w}_0)$$

Update $w_{k+1} \leftarrow \frac{1}{T}\sum_{t=0}^{T-1} \tilde{w}_t$.

Today we will operate under the following assumption.

Assumption (VR): Each $f_i : \mathbb{R}^d \to \mathbb{R}$ is convex and L-smooth with $f^* = \min f = f(w^*)$.

Last time we saw that

$$\mathbb{E}\,\|g(w, i; \tilde{w}) - \nabla f(w)\|^2 \le L^2 \|w - \tilde{w}\|^2.$$

The issue with this is that we can't control the distance $\|w - \tilde{w}\|$ without additional assumptions. Let's fix that.

Define $\Delta(w) := f(w) - f^*$.

Theorem (SVRG variance): Suppose that Assumption (VR) holds. Then,

$$\mathbb{E}_i \|g(w, i; \tilde{w})\|^2 \le 4L(\Delta(w) + \Delta(\tilde{w})) \quad \forall w, \tilde{w} \in \mathbb{R}^d.$$

$\mathbb{E}_i \|g(w, i; \tilde{w}) - \nabla f(w)\|^2$

Proof: Define

$a_i = \nabla f_i(w) - \nabla f_i(w^*)$ and $b_i = \nabla f_i(\tilde{w}) - \nabla f_i(w^*)$

(Might be nonzero)

We can write $g(w, i; \tilde{w}) = a_i - (b_i - \mathbb{E} b_i)$.

Hence

$$\mathbb{E}_i \|g(w,i;\tilde{w})\|^2 \le 2\mathbb{E}_i\|a_i\|^2 + 2\mathbb{E}_i\|b_i - \mathbb{E}b_i\|^2$$

why? → $\le 2\mathbb{E}_i\|a_i\|^2 + 2\mathbb{E}_i\|b_i\|^2$.

$\mathbb{E}b_i \in \arg\min_c \mathbb{E}\|b_i - c\|^2$

Hence, it suffices to show

$$\mathbb{E}_i\|\nabla f_i(w) - \nabla f_i(w^*)\|^2 \le 2L\Delta(w) \quad \forall w \in \mathbb{R}^d$$

To do so define

$$\varphi_i(w) = f_i(w) - f_i(w^*) - \langle \nabla f_i(w^*), w - w^* \rangle.$$

By convexity we have $\varphi_i(w) \ge 0$. Further, $\nabla\varphi_i(w) = \nabla f_i(w) - \nabla f_i(w^*)$. The descent lemma gives

$$0 \le \varphi_i\left(w - \tfrac{1}{L}\nabla\varphi_i(w)\right) \le \varphi_i(w) - \frac{1}{2L}\|\nabla\varphi_i(w)\|^2$$

Rearranging and taking expectations,

$\mathbb{E}_i \|\nabla f_i(w) - \nabla f_i(w^*)\|^2 \le 2L\,(f(w) - f(w^*))$,

which finishes the proof. □

To show that (SVRG) contracts the function gap, we need one extra assumption.

Assumption (aG). Suppose there is $\mu > 0$ such that for all $w \in \mathbb{R}^d$,

$$\frac{\mu}{2}\|w - w^*\|^2 \le f(w) - f^*.$$

Define $\mathcal{F}_{k,t}$ ($\mathcal{F}_k = \mathcal{F}_{k,0}$) to be the history just before index j_t is drawn in epoch k. (Outer iteration)

Lemma (one-step): For every epoch k and inner iteration t,

$$\mathbb{E}[\|\tilde{w}_{t+1} - w^*\|^2 \mid \mathcal{F}_{k,t}] \le \|\tilde{w}_t - w^*\|^2 - 2\eta(1 - 2L\eta)\Delta(\tilde{w}_t) + 4L\eta^2\Delta(w_k).$$

Proof: Expanding the square

$$\mathbb{E}\left[\|\tilde{w}_t - \eta g(\tilde{w}_t, i; w_k) - w^*\|^2 \mid \mathcal{F}_{k,i}\right]$$

$$= \|\tilde{w}_t - w^*\|^2 - 2\eta\langle \nabla f(\tilde{w}_t), \tilde{w}_t - w^*\rangle + \eta^2 \mathbb{E}\left[\|g_t\|^2 \mid \mathcal{F}_{k,t}\right]$$

$$\leq \|\tilde{w}_t - w^*\|^2 - 2\eta\,\Delta(\tilde{w}_t) + 4L\eta^2\left[\Delta(\tilde{w}_t) + \Delta(w_k)\right]$$

Convexity; Theorem (SVRG). □

Theorem (SVRG linear converge). Suppose Assumptions (VR) and (SG) hold, $\eta < \frac{1}{2L}$, $T \geq 1$, and set

$$\rho = \frac{\frac{1}{\mu\eta T} + 2L\eta}{1 - 2L\eta}.$$

$\rho \in (0,1)$ for suitable η and T.

Then,

$$\mathbb{E}\left[f(w_{k+1}) - f^* \mid \mathcal{F}_k\right] \leq \rho\left(f(w_k) - f^*\right).$$

Hence, if $\rho < 1$,

$$\mathbb{E}\, f(w_k) - f^* \leq \rho^k\left(f(w_0) - f^*\right) \quad \forall k.$$

Proof: By Lemma (one-step) and the tower property, we have

$$2\eta(1-2L\eta)\sum_{t=0}^{T-1}\mathbb{E}[\Delta(\tilde{w}_t)\mid\mathcal{F}_k]$$

(Telescope) $$\le \|w_k - w^*\|^2 - \mathbb{E}[\|\tilde{w}_T - w^*\|^2 \mid \mathcal{F}_k] + 4L\eta^2 T\Delta(w_k)$$

$$\le \|w_k - w^*\|^2 + 4L\eta^2 T\Delta(w_k).$$

Rearranging and taking an average

$$\mathbb{E}[\Delta(w_{k+1})\mid\mathcal{F}_k] \overset{\text{Jensen's}}{\le} \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\Delta(\tilde{w}_t)\mid\mathcal{F}_k]$$

$$\le \frac{\|w_k - w^*\|^2}{2\eta(1-2L\eta)T} + \frac{2L\eta\Delta(w_k)}{(1-2L\eta)}$$

$$\le \left(\frac{1}{\mu\eta(1-2L\eta)T} + \frac{2L\eta}{(1-2L\eta)}\right)\Delta(w_k)$$

$$= \rho\,\Delta(w_k).$$

□

Corollary. Take $\eta = \frac{1}{10L}$ and $T = \lceil 50 \cdot \frac{L}{\mu} \rceil$. Then

$$\mathbb{E}[f(w_k) - f^*] \leq 2^{-k}(f(w_0) - f^*).$$

So how much do we gain? we need to measure this in terms of component-gradient evaluations (i.e., $\nabla f_i(w)$). For GD, we showed in Lecture 2 that we needed a grant total of

$$O\left(n \frac{L}{\mu} \log\left(\frac{f(w_0) - f^*}{\varepsilon}\right)\right)$$

↑ Each iteration requires n evaluations.

For (SVRG) we need n evaluations at the start of the epoch and $\frac{L}{\mu}$ evaluations for the rest of that epoch so, we need

$$O\left(\left(n + \frac{L}{\mu}\right) \log\left(\frac{f(w_0) - f^*}{\varepsilon}\right)\right).$$

↑ Super nice improvement. Think of the case $n \approx L/\mu$.

Remarks

- The overall function f might have a better L-smoothness constant, so it is not always a fair comparison with GD.

- Hardware plays a big role in wall-clock time, vectorization and parallelization can make computing a gradient cheap. It depends on the scale.

- There are many natural follow-ups, what if we don't get a full gradient, but a better approximation? We will not cover any of that.